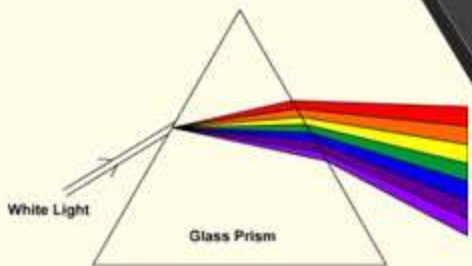


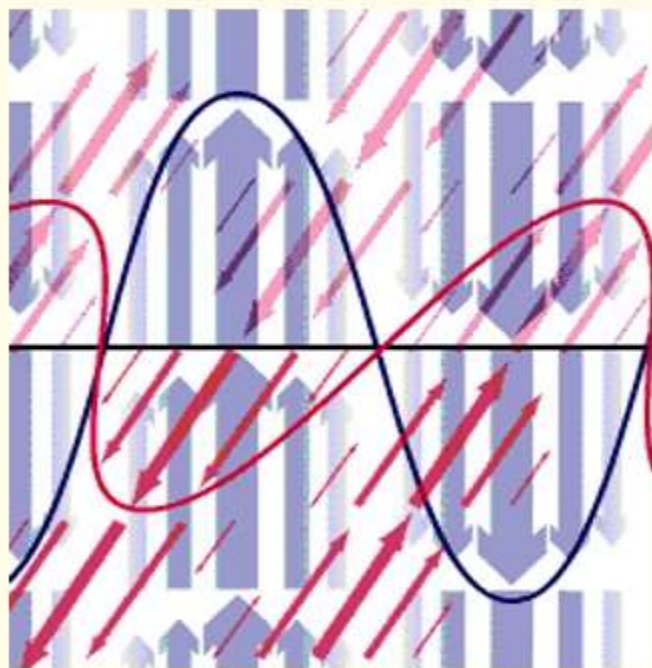
PHYSICS

OPTICS REFRACTION OF LIGHT

MODULE - 6



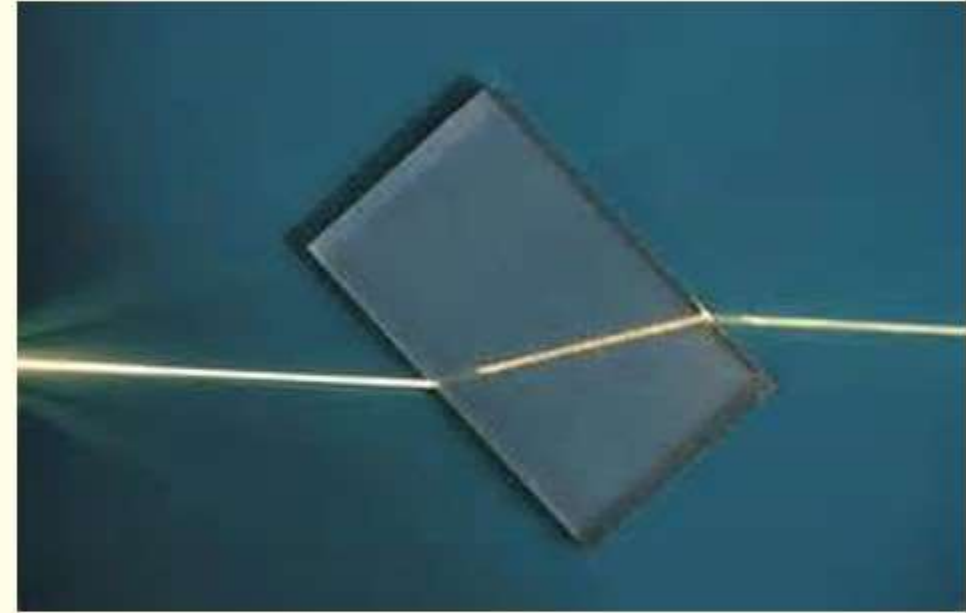
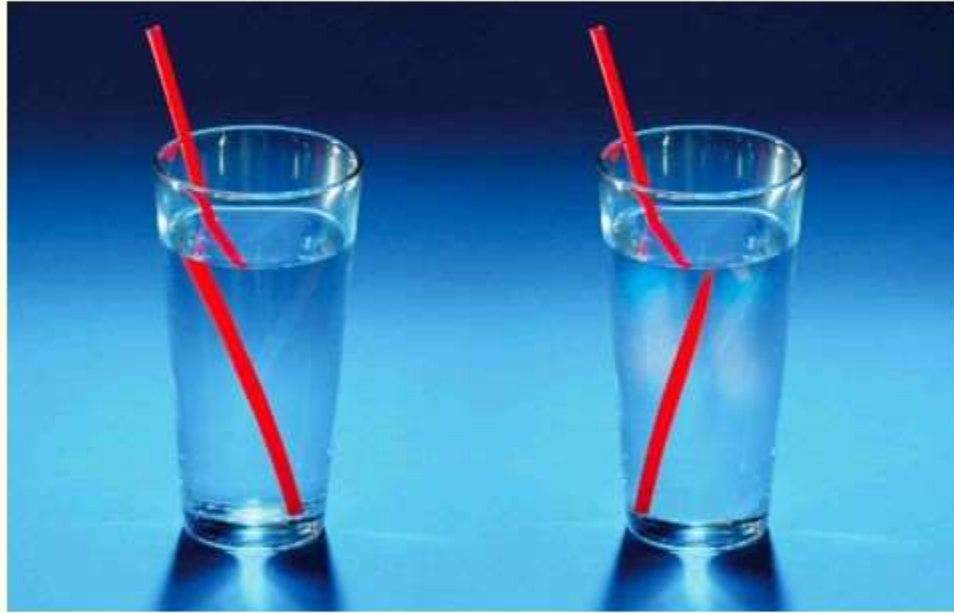
LIGHT



- Light is a form of energy that is propagated by electromagnetic waves.
- The magnetic and electric field is perpendicular to each other.
- It is a form of energy that reaches Earth enabling us to see.
- The scientific study of light is called OPTICS.

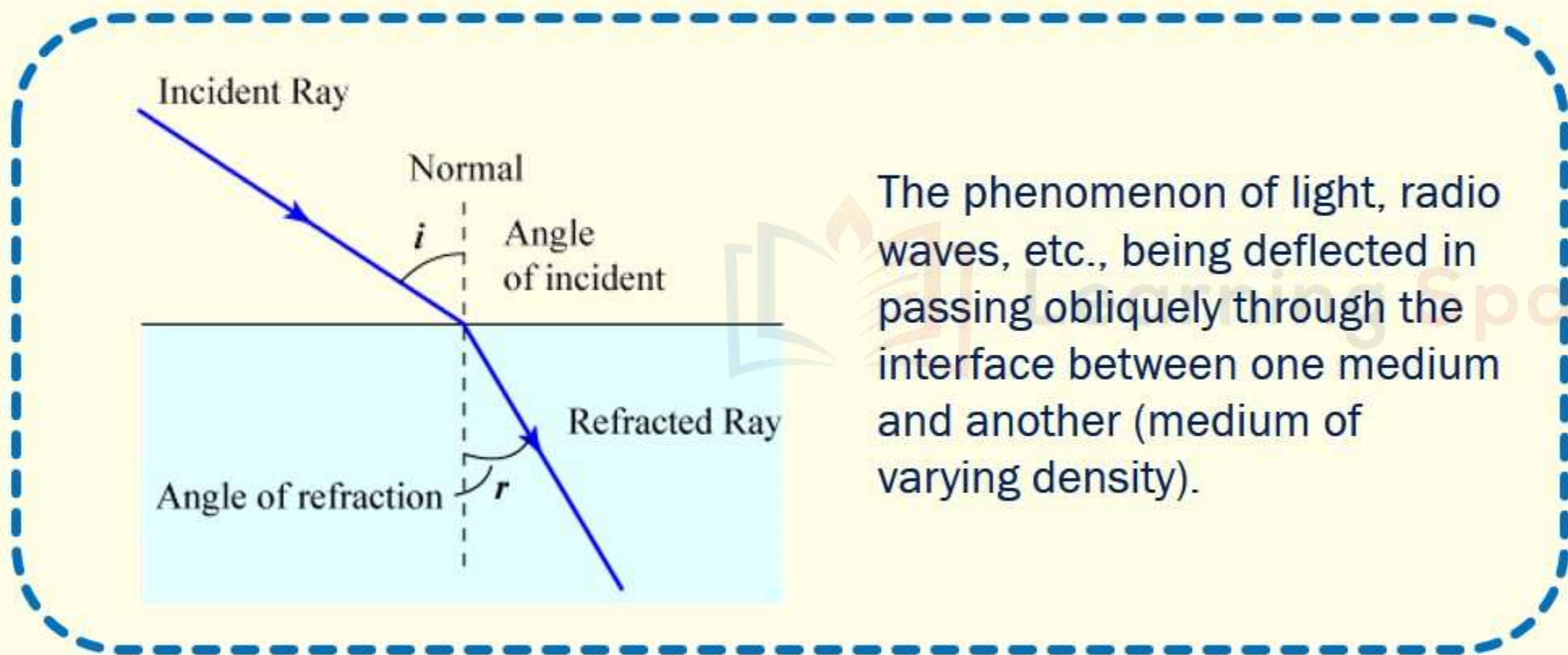
Electromagnetic wave is formed when an electric field (blue arrows) couples with a magnetic field (red arrows).

HAVE YOU BEEN THROUGH THIS OBSERVATION ?



What we learnt – light travels in a straight line
What we see – light can bend its path

REFRACTION OF LIGHT



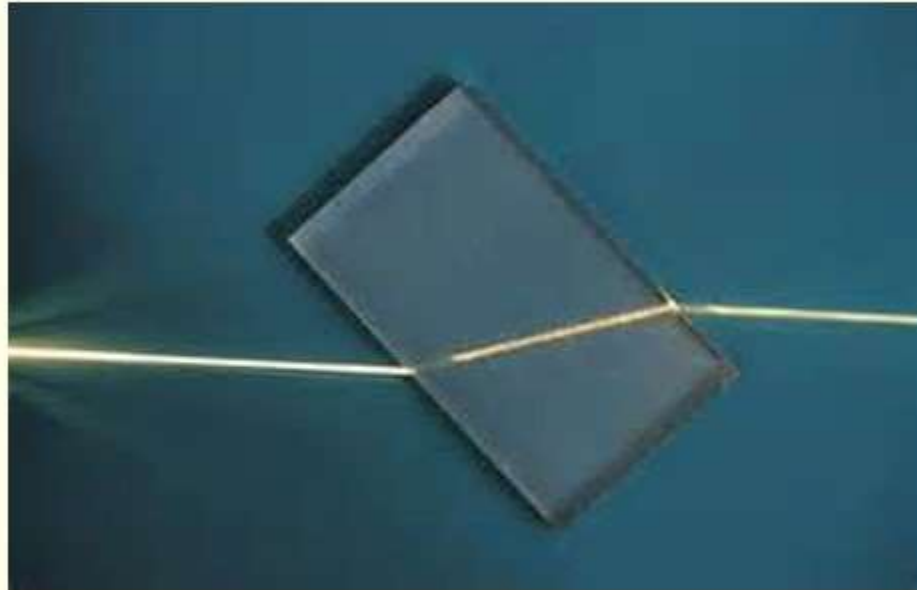
The phenomenon of light, radio waves, etc., being deflected in passing obliquely through the interface between one medium and another (medium of varying density).

There will be no bending if the light ray travels parallel to the normal





WHY LIGHT BENDS ITS PATH?

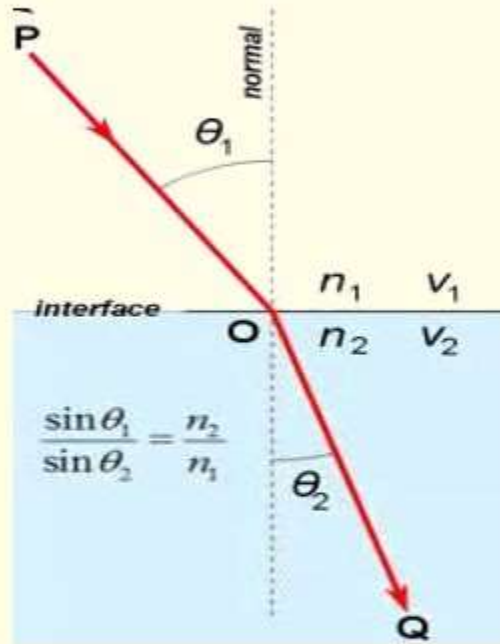


- LIGHT travels at different speeds through different medium because some mediums are dense while some mediums are rare (less dense).
- The more optically dense a medium is the more it can bend the light inwards because the speed of the light decreases on entering the medium.
- In an optically rare medium, the light bends outwards because the speed of the light increases on entering the medium.

A lateral shift in the direction of incident light is the lateral deviation



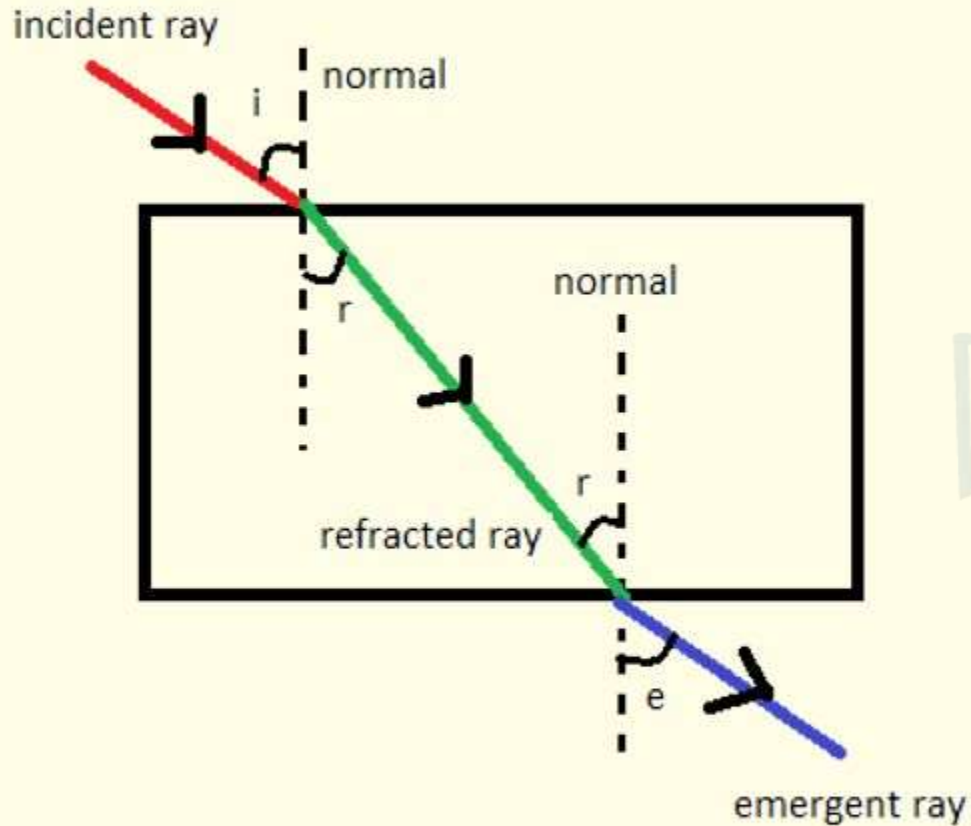
WHAT LAWS DOES REFRACTION FOLLOW ?



- Incident ray, reflected ray, refracted ray and the normal of the system lie in the same plane.
- Incident ray, coming from one medium to the boundary of another medium, is refracted with a rule derived from a physicist Willebrord Snellius. He found that there is a constant relation between the angle of incident ray and angle of refracted ray.
- The ratio between the sine of angle of incidence to sine of angle refraction is a constant called as the refractive index.



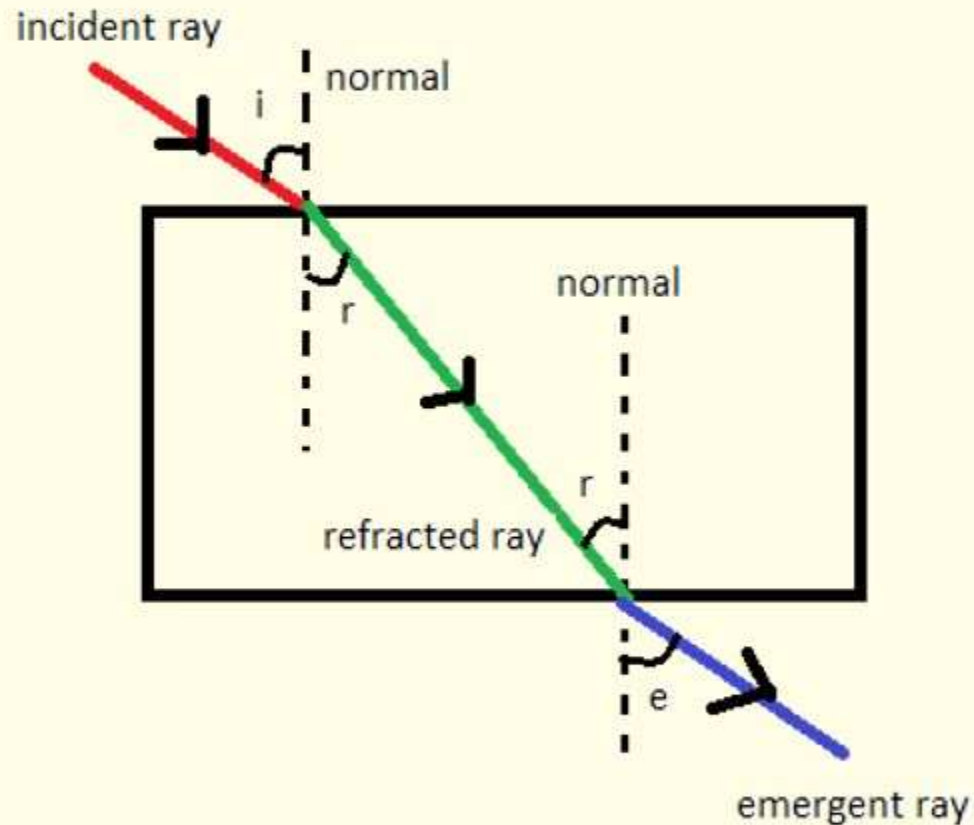
THE SNELL'S LAW – IN A SIMPLE WAY !



- When the incident ray enters an optically denser medium, the ray bends towards the normal; i.e. the sine of angle of refraction becomes less.
- On the other hand, when the incident ray comes out from an optically dense medium to an optically rarer medium, the opposite happens.
- In that case, the refracted ray bends away from the normal and the sine of angle of refraction is more.
- This explains the Snell's Law.



REFRACTION & REFRACTIVE INDEX - RELATION



- The bending of light from one medium to another depends upon the refractive index of the two medium and the angle of incidence on the line of separation of the two medium.
- A higher refractive index means light will bend in that medium more than on a medium of lower refractive index.
- Vacuum has a refractive index of 1 so it bends the light least but crown glass used in optical lenses has a refractive index of ~ 1.5 which means light will bend on entering it.



WHAT MAKES THE DIFFERENCE ?

REFLECTION

- Light bounces off the reflecting surface.
- All wavelengths are affected equally .

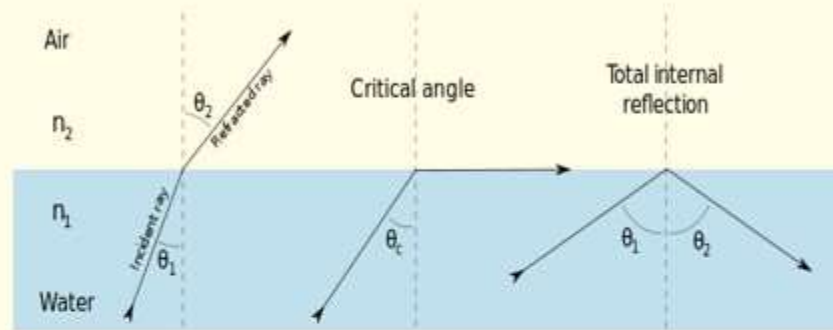


REFRACTION

- Light is bent from its original direction at the refracting surface.
- Different wavelengths are affected differently



COMMON OBSERVATION OF REFRACTION



TOTAL INTERNAL REFLECTION

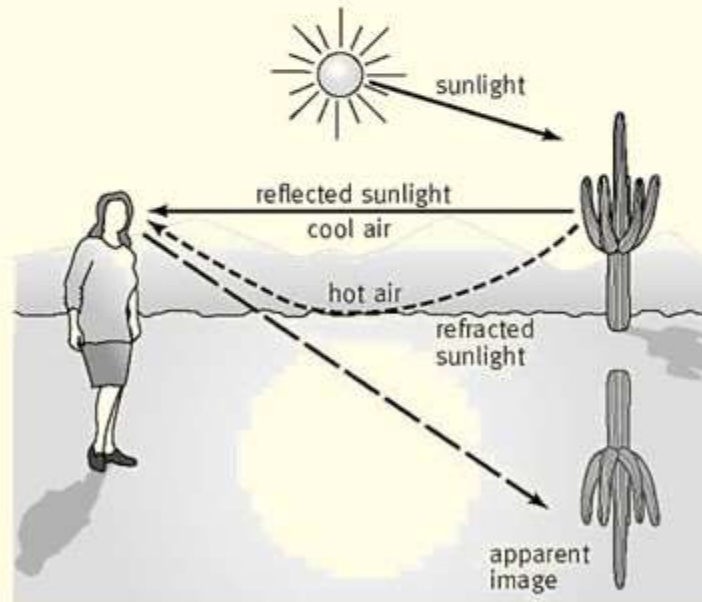
- Total internal reflection is the phenomenon which occurs when a light ray strikes a medium boundary **at an angle larger than a particular critical angle** with respect to the normal to the surface.
- When a light ray travels from denser to rarer medium and **the incident angle is greater than the critical angle**, the ray cannot pass through and is entirely reflected.
- Applications - used in optical fibres which is used in endoscopy for imaging internal organs and telecommunications
- Diamond's cut enhance the total internal reflection to make it extremely shiny.

COMMON OBSERVATION OF REFRACTION



MIRAGE

- On a hot and sunny day, a mirage is quite common which gives an onlooker a false image of moist ground ahead.
- A mirage is an optical illusion which is generated when the layer of air near the earth's surface is lighter than the upper layers of the air.
- Due to the variation of the optical densities of air layers, light from an object undergoes a series of internal reflections and bend upwards.
- It thus gives an inverted image of the seen object, to a distant onlooker it feels it's a reflection in the water.



COMMON OBSERVATION OF REFRACTION

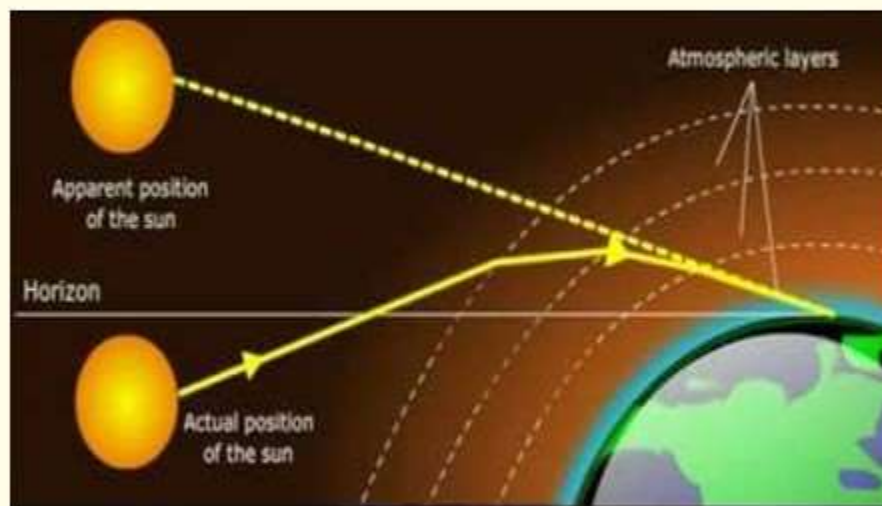
TWINKLING OF STARS



- The twinkling of stars is due to atmospheric refraction of star-light.
- The atmosphere is made of several layers of air with different refractive indices which keep on changing continuously.
- Due to this path of light rays from the star coming through the layers of the atmosphere keeps on changing their path continuously.
- As a consequence the number of rays entering the pupil of eye goes on changing with time and the stars appear twinkling.
- The longer the light travels to reach Earth, the more it will twinkle and the shorter the path the more it will look as if it's steady.
- This explains why Pole Star does not seem to twinkle.

COMMON OBSERVATION OF REFRACTION

APPARENT POSITION OF SUN



- The atmospheric refraction also affects the position of the Sun as we perceive it. We cannot see the real position of the sun in the sky but a refracted image.
- This is because the density of the atmospheric layer varies with altitude.
- **Refractive index of the atmospheric layer decreases as from bottom to top.**
- The light rays travel from rarer to denser layer starting from the Sun and continuously get bent more towards the normal.

DO YOU KNOW ?

We actually see the sunrise about 2 minutes before the sun is actually at that perceived position. Also, the sunset that we see is actually of a sun that has, in reality, already set.

REFRACTION OF LIGHT IN A PRISM

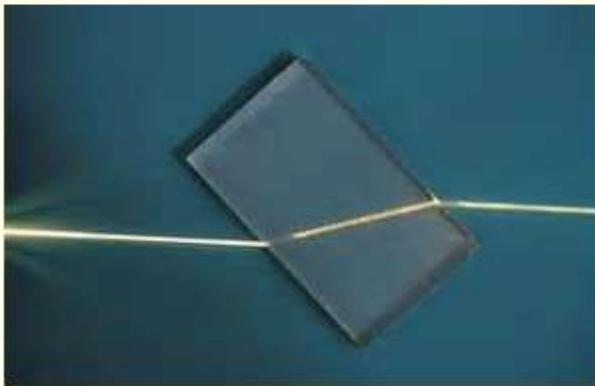
DISPERSION



- Visible light, also known as white light, consists of a collection of component colors.
- Upon passage through the prism, the white light is separated into its component colors - red, orange, yellow, green, blue and violet.
- The separation of visible light into its different colors is known as dispersion.
- Each color is characteristic of a distinct wave frequency; and different frequencies of light waves will bend varying amounts upon passage through a prism because of their difference in their speed in a particular medium.
- While in vacuum or air, they all have a similar speed thus they combine to give white colour, but the speed of propagation changes in denser medium like a glass prism.



WHY CAN A PRISM DISPERSE LIGHT BUT A GLASS SLAB CANNOT ?



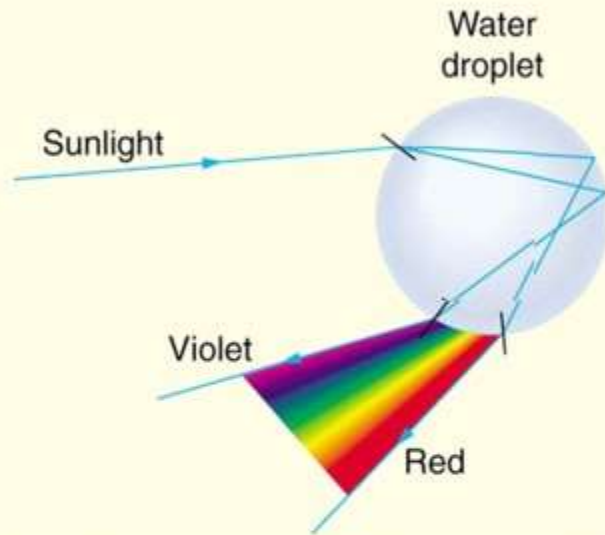
- In the case of a glass slab, the opposite sides are parallel and therefore different colors emerge parallel to each other and are seen simultaneously.
- Therefore, dispersion doesn't occur in a glass slab.
- In a prism, the opposite sides are not parallel to each other so all the colors of light are refracted by different amounts by different angles.
- The violet colour is the most deviated and the red colour is least deviated.

The angle between the incident ray and the emergent ray is called the angle of deviation.



NATURAL OCCURANCE OF DISPERSION

RAINBOW IN THE SKY



- A collection of suspended water droplets in the atmosphere serves as a refractor of light. The water represents a medium with a different optical density than the surrounding air.
- When sunlight is incident at particular angle, light refracts into the droplet, and then experiences internal reflection, and then refracts out of the droplet.
- On coming out of the droplet, the light ray is dispersed and bent downward towards an observer on earth's surface.
- The net effect of the vast array of droplets is that a circular arc of **ROYGBIV** is seen across the sky. (Red remains at the top of the rainbow).

its angle of deviation from the original sun rays is approximately 40 degrees





A DOUBLE RAINBOW - WHY ?

SECONDARY RAINBOW



- A secondary rainbow happens when the Sunlight is reflected twice in the raindrops.
- A primary rainbow occurs when the angle of deviation of the sunlight is 40- 42 degrees, but in a Secondary rainbow, the angle increases to 51 degrees.
- The secondary rainbow has the colour reversed than the primary rainbow, i.e. VIBGYOR. It is twice as wide and has the intensity reduced to $1/10^{\text{th}}$ of the primary one.

Rainbows cannot be seen at mid-day because the sun light cannot make the required angles, it is formed in the afternoons because if the Sun's lower position



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